

406. On the Curves which satisfy given Conditions (continued)

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[406] On the Curves which satisfy given Conditions.

ARTICLE NOS. 26 TO 41.

26. A single condition (Σ) imposed upon a conic has two representative numbers or simply representatives, (α, β) ; viz. if (4Σ) be an arbitrary system of four conditions and (μ, ν) the characteristics of (4Σ) , then the number of the conics which satisfy the five conditions $(\Sigma, 4\Sigma)$ is $= \alpha\mu + \beta\nu$.

27. As an instance of the use of the characteristics, if $\Sigma, \Sigma', \Sigma'', \Sigma''', \Sigma''''$ be any five independent conditions, and $(\alpha, \beta), \dots, (\alpha''''', \beta''''')$ the representatives of these conditions respectively, then the number of the conics which satisfy the five conditions $(\Sigma, \Sigma', \Sigma'', \Sigma''', \Sigma''''')$ is

$$= (1, 2, 4, 4, 2, 1 \mid \alpha, \beta)(\alpha', \beta')(\alpha'', \beta'')(\alpha''', \beta''')(\alpha''''', \beta'''''),$$

viz. this notation stands for $1\alpha\alpha'\alpha''\alpha'''\alpha'''' + 2\Sigma\alpha\alpha'\alpha''\alpha'''\beta'''' \dots + 1\beta\beta'\beta''\beta'''\beta''''$.

28. In particular if Σ be the condition that a conic shall touch a given curve of the order m and class n , then the representatives of this condition are (n, m) . Hence the number of the conics which touch each of five given curves $(m, n), \dots$ is

$$= (1, 2, 4, 4, 2, 1 \mid n, m)(n', m')(n'', m'')(n''', m''')(n''''', m''''').$$

29. A system of conics (4Σ) having the characteristics (μ, ν) , contains

- $2\nu - \mu$ line-pairs, that is, conics each of them a pair of lines; and
- $2\mu - \nu$ point-pairs, that is, conics each of them a pair of points (*coniques infiniment aplaties*).

30. I stop to further explain these notions of the line-pair and the point-pair—and also the notion of the ‘line-pair-point.’

A conic is a curve of the second order and second class; quà curve of the second order it may degenerate into a pair of lines, or line-pair (but the class is then $= 0$)—quà curve of the second class it may degenerate into a pair of points, or point-pair (but the order is then $= 0$). The two lines of a line-pair may be coincident and we have then a coincident line-pair; such a line-pair (it must I think be postulated) ordinarily arises, not from a line-pair the two lines of which become coincident but from a proper conic, flattening by the gradual diminution of its

conjugate axis while its transverse axis remains constant or approaches a limit different from zero—the conic thus tends (not to an indefinitely extended but) to a terminated line C)—in other words, the tangents of the conic become more and more nearly lines through two fixed points, the terminations of the terminated line; and these terminating points which continue to exist up to the instant when the conjugate axis takes its limiting value $= 0$, are regarded as still existing at this instant, and the coincident line-pair as being in fact the point-pair formed by the two terminating points. Similarly the two points of a point-pair may be coincident, and we have then a coincident point-pair; such a point-pair (it must in like manner be postulated) ordinarily arises, not from a point-pair the two points of which become coincident, but from a proper conic sharpening itself to coincide with its asymptotes, and so becoming ultimately a pair of lines through the coincident point-pair; and the coincident point-pair is regarded as being in fact the line-pair formed by some two lines through the coincident point-pair.

31. In accordance with the foregoing notions we may with propriety, and it will in the sequel be found convenient to speak of a point-pair as a line terminated by two points on this line, and similarly to speak of a line-pair as a point terminated (that is, the pencil of lines through the point is terminated) by two lines through the point.

32. If in a point-pair thus considered as a line terminated by two points the two points become coincident (the line continuing to exist as a definite line), or, what is the same thing, if in a line-pair thus considered as a point terminated by two lines, the two lines become coincident (the point continuing to exist as a definite point), we have a “line-pair-point;” viz. this is at once a coincident line-pair and a coincident point-pair; it may also be regarded as the limit of a conic the axes of which, and the ratio of the conjugate to the transverse axis, all ultimately vanish: it may be described as a line terminated each way at a point thereof, or as a point terminated each way at a line through it. The notion of a line-pair-point first presents itself in Zeuthen’s researches, as will presently appear; but it may be noticed here that line-pair-points, and these the same line-pair-points, may present themselves among the $2\nu - \mu$ line-pairs, and among the $2\mu - \nu$ point-pairs of the system of conics 4Σ .

33. Returning to the foregoing theory of characteristics, I remark that the fundamental notion may be taken to be, not the characteristics (μ, ν) of the conics which satisfy four conditions, but in every case the number of the conics which satisfy five conditions. Thus for the conics not subjected to any condition, we may consider the symbols

$$(::::), (::: /), (: : //), (: : ///), (////),$$

denoting the number of the conics which pass through five given points, or which pass through four given points and touch a given line, &c. . . ., or which touch five given lines; these numbers are respectively

$$= 1, 2, 4, 4, 2, 1.$$

So for the conics which satisfy a given condition Σ , or two conditions 2Σ , . . . , or five conditions 5Σ , we have respectively the numbers

$$\begin{array}{ll} \Sigma, & (: : \cdot) \Sigma, \quad (\cdot \cdot /) \Sigma, \quad (: / \cdot) \Sigma, \quad (\cdot ///) \Sigma, \quad (////) \Sigma, \\ 2\Sigma, & (: : \cdot) \Sigma, \quad (\cdot /) \Sigma, \quad (: / \cdot) \Sigma, \quad (///) \Sigma, \\ 3\Sigma, & (: : \cdot) \Sigma, \quad (\cdot /) \Sigma, \quad (//) \Sigma, \\ 4\Sigma, & (\cdot) \Sigma, \quad (/) \Sigma, \\ 5\Sigma, & (\text{single number}). \end{array}$$

where the Σ , 2Σ , &c. belong to the symbols which follow: read $(\Sigma ::::)$, $(\Sigma :: : /)$, &c., or, as we may for shortness represent them,

$$\mu'''' , \nu'''' , \rho'''' , \sigma'''' , \tau'''' ,$$

viz. the single condition Σ has the five characteristics $(\mu'''' , \dots \tau'''')$, ...; the four conditions 4Σ , the characteristics (μ, ν) as in the original theory; and the five conditions 5Σ a single characteristic μ_0 .

34. We thus see the origin of the notion of the representatives (α, β) of a single condition Σ ; for considering the arbitrary four conditions 4Σ , the characteristics whereof are (μ, ν) , and assuming that the single characteristic, or number of the conics $(\Sigma, 4\Sigma)$, is $= \alpha\mu + \beta\nu$, and taking for (4Σ) successively the conditions

$$(\vdots\vdots), (\vdots\vdots /), (\vdots\vdots //), (\vdots\vdots ///), (//////),$$

having respectively the characteristics

$$(1, 2), (2, 4), (4, 4), (4, 2), (2, 1),$$

we have

$$\begin{aligned}\mu'''' &= 1\alpha + 2\beta, \\ \nu'''' &= 2\alpha + 4\beta, \\ \rho'''' &= 4\alpha + 4\beta, \\ \sigma'''' &= 4\alpha + 2\beta, \\ \tau'''' &= 2\alpha + 1\beta,\end{aligned}$$

that is, the characteristics $(\mu'''' , \nu'''' , \rho'''' , \sigma'''' , \tau'''')$ of a single condition Σ are not independent, but are representable as above by means of two independent quantities (α, β) ; or, what is the same thing, we have

$$\nu'''' = 2\mu'''' , \quad \rho'''' = \mu'''' + \sigma'''' , \quad \tau'''' = \frac{1}{2}\nu'''' ,$$

which being satisfied, the representatives (α, β) are given by

$$\alpha = \frac{1}{2}(2\tau'''' - \nu''''), \quad \beta = \frac{1}{2}(2\mu'''' - \tau'''').$$

35. I find that a like property exists as to the characteristics $(\mu'', \nu'', \rho'', \sigma'')$ of the two conditions 2Σ , viz. these are not independent but are connected by a single linear relation,

$$2\mu'' - 2\nu'' + \rho'' - 2\sigma'' = 0.$$

This may be proved in the case where the conditions 2Σ are two separate conditions (Σ, Σ') ; viz. let the representatives of these be (α, β) , (α', β') respectively, then

combining with them the three arbitrary conditions Σ'' , Σ''' , Σ'''' having respectively the representatives (α'', β'') , (α''', β''') , (α'''' , β''''), we have the general equation

$$(\Sigma, \Sigma', \Sigma'', \Sigma''', \Sigma'''') = (1, 2, 4, 4, 2, 1 \mid \alpha, \beta)(\alpha', \beta')(\alpha'', \beta'')(\alpha''', \beta''')(\alpha'''' , \beta'''');$$

taking herein

$$(\Sigma'', \Sigma''', \Sigma'''') = (\vdots\vdots), (\vdots\vdots /), (\vdots\vdots //), (\vdots\vdots ///)$$

successively, and observing that the representatives of $(\vdots\vdots \cdot)$ are $(1, 0)$ and those of $(/)$ are $(0, 1)$, we thus obtain for $(\mu'', \nu'', \rho'', \sigma'')$, characteristics of (Σ, Σ') , the values

$$\begin{aligned}\mu'' &= (1, 2, 4 \mid \alpha, \beta)(\alpha', \beta'), \\ \nu'' &= (2, 4, 4 \mid \alpha, \beta)(\alpha', \beta'), \\ \rho'' &= (4, 4, 2 \mid \alpha, \beta)(\alpha', \beta'), \\ \sigma'' &= (4, 2, 1 \mid \alpha, \beta)(\alpha', \beta'),\end{aligned}$$

(viz. $\mu'' = 1\alpha\alpha' + 2(\alpha\beta' + \alpha'\beta) + 4\beta\beta'$, &c.), and these values give identically

$$2\mu'' - 2\nu'' + \rho'' - 2\sigma'' = 0,$$

which is the foregoing equation. And I assume that the theorem extends to the case of two inseparable conditions 2Σ , but in this case I do not even know where the proof is to be sought for.

The characteristics (μ', ν', ρ') of the three conditions 3Σ are in general independent.

36. It has been mentioned that if (α, β) are the representatives of the condition Σ , and (μ, ν) the characteristics of the conditions 4Σ , then

$$(\Sigma, 4\Sigma) = \alpha\mu + \beta\nu;$$

this is the most convenient form of the theorem, but as (α, β) are known functions of the characteristics $(\mu''''', \nu''''', \rho''''', \sigma''''', \tau''''')$ of the condition Σ , the equation is in effect an expression for $(\Sigma, 4\Sigma)$ in terms of the characteristics of Σ and 4Σ respectively.

There is, similarly, an expression for $(2\Sigma, 3\Sigma)$ in terms of the characteristics $(\mu', \nu', \rho', \sigma')$ of 3Σ (satisfying the relation $\mu' - 2\nu' + \frac{1}{2}\rho' - \sigma' = 0$) and the characteristics (μ, ν, ρ) of 2Σ , viz. we have

$$(2\Sigma, 3\Sigma) = \mu(-\frac{1}{4}\rho' + \frac{1}{2}\sigma') + \nu(\frac{1}{2}\mu' - \frac{1}{4}\rho') + \rho(\frac{1}{2}\mu' - \frac{1}{2}\nu' + \frac{1}{4}\rho').$$

This may be easily proved in the case where the conditions 2Σ are two separable conditions Σ, Σ' having the representatives $(\alpha, \beta), (\alpha', \beta')$ respectively, and the conditions 3Σ three separable conditions $\Sigma'', \Sigma''', \Sigma''''$ having the representatives $(\alpha'', \beta''), (\alpha''', \beta'''), (\alpha'''', \beta''')$ respectively; we have, in fact,

$$\begin{aligned} \mu' &= (1, 2, 4 \mid \alpha', \beta'), & \mu &= (1, 2, 4, 4 \mid \alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta'''), \\ \nu' &= (2, 4, 4 \mid \alpha', \beta'), & \nu &= (2, 4, 4, 2 \mid \alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta'''), \\ \rho' &= (4, 4, 2 \mid \alpha', \beta'), & \rho &= (4, 4, 2, 1 \mid \alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta'''), \\ \sigma' &= (4, 2, 1 \mid \alpha', \beta'); \end{aligned}$$

and with these values the function

$$(\Sigma, \Sigma', \Sigma'', \Sigma''', \Sigma''') = (1, 2, 4, 4, 2, 1 \mid \alpha, \beta)(\alpha', \beta')(\alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta''')$$

is found to be expressible as above in terms of $(\mu, \nu, \rho), (\mu', \nu', \rho', \sigma')$; but I do not know how to conduct the proof for the inseparable conditions 2Σ and 3Σ .

37. It may be remarked by way of verification that writing successively

$$(3\Sigma) = (::: \cdot), (:: / :), (: //), (: ///),$$

that is,

$$(\mu, \nu, \rho) = (1, 2, 4), (2, 4, 4), (4, 4, 2), (4, 2, 1),$$

we have in the first case

$$(2\Sigma, \cdot) = -\frac{1}{4}\rho' + \frac{1}{2}\sigma' + 2\mu' - \nu' + \rho' - 2\sigma' = \mu',$$

and similarly in the other three cases,

$$(2\Sigma, /) = \nu', \quad (2\Sigma, //) = \rho', \quad (2\Sigma, ///) = \sigma'.$$

38. Let (μ, ν, ρ, σ) be the characteristics of 2Σ , $(\mu - 2\nu + \frac{1}{2}\rho - \sigma = 0)$, and $(\mu', \nu', \rho', \sigma')$ the characteristics of $2\Sigma'$, $(\mu' - 2\nu' + \frac{1}{2}\rho' - \sigma' = 0)$. Then in the formula for $(2\Sigma, 3\Sigma)$, writing successively for 3Σ

$$\begin{aligned}(3\Sigma) &= (2\Sigma'), & \text{characteristics } (\mu', \nu', \rho'), \\ (3\Sigma) &= (2\Sigma', 1), & \text{characteristics } (\nu', \rho', \sigma'),\end{aligned}$$

we obtain expressions for the characteristics $(2\Sigma, 2\Sigma')$, $(2\Sigma, 2\Sigma', 1)$, which are consequently

$$\begin{aligned}\mu(2\Sigma, 2\Sigma') &= \mu(-\tfrac{1}{4}\rho' + \tfrac{1}{2}\sigma') + \nu(\tfrac{1}{2}\mu' - \tfrac{1}{4}\rho') + \rho(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho'), \\ \nu(2\Sigma, 2\Sigma') &= \mu(-\tfrac{1}{4}\rho' + \tfrac{1}{4}\sigma') + \nu(\tfrac{1}{2}\mu' - \tfrac{1}{4}\rho' + \tfrac{1}{4}\sigma') + \rho(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho' - \tfrac{1}{4}\sigma'), \\ \rho(2\Sigma, 2\Sigma') &= \mu(0) + \nu(\tfrac{1}{2}\mu' - \tfrac{1}{4}\rho' + \tfrac{1}{4}\sigma') + \rho(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho' - \tfrac{1}{4}\sigma'), \\ \sigma(2\Sigma, 2\Sigma') &= \mu(0) + \nu(0) + \rho(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho' - \tfrac{1}{4}\sigma'),\end{aligned}$$

and similarly

$$\begin{aligned}\mu(2\Sigma, 2\Sigma', 1) &= \mu(-\tfrac{1}{4}\rho' + \tfrac{1}{2}\sigma') + \nu(\tfrac{1}{2}\mu' - \tfrac{1}{4}\rho') + \rho(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho'), \\ \nu(2\Sigma, 2\Sigma', 1) &= \mu(0) + \nu(\tfrac{1}{2}\mu' - \tfrac{1}{4}\rho' + \tfrac{1}{4}\sigma') + \rho(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho' - \tfrac{1}{4}\sigma'), \\ \rho(2\Sigma, 2\Sigma', 1) &= \mu(0) + \nu(0) + \rho(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho' - \tfrac{1}{4}\sigma').\end{aligned}$$

39. If in these formulae for $(2\Sigma, 2\Sigma')$, $(2\Sigma, 2\Sigma', 1)$, we write $(\mu, \nu, \rho, \sigma) = (\mu', \nu', \rho', \sigma')$, or, what is the same thing, write $\Sigma = \Sigma'$, then observing that these characteristics satisfy the relation $\mu - 2\nu + \frac{1}{2}\rho - \sigma = 0$, the formulae become

$$\begin{aligned}\mu(4\Sigma) &= \mu(-\tfrac{1}{4}\rho + \tfrac{1}{2}\sigma) + \nu(\tfrac{1}{2}\mu - \tfrac{1}{4}\rho) + \rho(\tfrac{1}{2}\mu - \tfrac{1}{2}\nu + \tfrac{1}{4}\rho), \\ \nu(4\Sigma) &= \mu(-\tfrac{1}{4}\rho + \tfrac{1}{4}\sigma) + \nu(\tfrac{1}{2}\mu - \tfrac{1}{4}\rho + \tfrac{1}{4}\sigma) + \rho(\tfrac{1}{2}\mu - \tfrac{1}{2}\nu + \tfrac{1}{4}\rho - \tfrac{1}{4}\sigma), \\ \rho(4\Sigma) &= \nu(\tfrac{1}{2}\mu - \tfrac{1}{4}\rho + \tfrac{1}{4}\sigma) + \rho(\tfrac{1}{2}\mu - \tfrac{1}{2}\nu + \tfrac{1}{4}\rho - \tfrac{1}{4}\sigma), \\ \sigma(4\Sigma) &= \rho(\tfrac{1}{2}\mu - \tfrac{1}{2}\nu + \tfrac{1}{4}\rho - \tfrac{1}{4}\sigma),\end{aligned}$$

which are therefore the characteristics of 4Σ in terms of those of 2Σ .

And for $(2\Sigma, 2\Sigma', 1)$, writing therein $(\mu, \nu, \rho) = (\nu', \rho', \sigma')$, that is, taking $2\Sigma = (2\Sigma', 1)$, we obtain the characteristics of $(4\Sigma, 1)$, which are consequently

$$\begin{aligned}\mu(4\Sigma, 1) &= \nu'(-\tfrac{1}{4}\rho' + \tfrac{1}{2}\sigma') + \rho'(\tfrac{1}{2}\mu' - \tfrac{1}{4}\rho') + \sigma'(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho'), \\ \nu(4\Sigma, 1) &= \rho'(\tfrac{1}{2}\mu' - \tfrac{1}{4}\rho' + \tfrac{1}{4}\sigma') + \sigma'(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho' - \tfrac{1}{4}\sigma'), \\ \rho(4\Sigma, 1) &= \sigma'(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho' - \tfrac{1}{4}\sigma').\end{aligned}$$

40. The characteristics of 5Σ , $(\Sigma, 4\Sigma) = \alpha\mu + \beta\nu$, are given in terms of the characteristics (μ, ν) of 4Σ and the representatives (α, β) of Σ . It would be interesting to obtain the characteristics of 6Σ in terms of those of 4Σ ; but I have not been able to effect this.

41. I remark that from the expression for the characteristics of 4Σ in terms of those of 2Σ , we may conversely express the characteristics of 2Σ in terms of those of 4Σ , and then the expression for $(\Sigma, 4\Sigma)$ may be written

$$(\Sigma, 4\Sigma) = \alpha\mu + \beta\nu = (a, b, c, d \mid \mu, \nu, \rho, \sigma),$$

where (a, b, c, d) are the characteristics of 2Σ . The same expression may be written

$$(\Sigma, 4\Sigma) = (a, b, c, d \mid \mu', \nu', \rho', \sigma'),$$

where $(\mu', \nu', \rho', \sigma')$ are the characteristics of $2\Sigma'$; and it would be interesting to find the relation between the characteristics (a, b, c, d) and (a', b', c', d') of 2Σ and $2\Sigma'$ respectively, such that the two expressions may be equivalent.

ARTICLE NOS. 42 TO 57.

42. In the case where the condition Σ is that of touching a given curve of the order m and class n , we have $(\alpha, \beta) = (n, m)$. Hence for two such conditions, touching two given curves (m, n) , (m', n') , the characteristics $(\mu'', \nu'', \rho'', \sigma'')$ are

$$\begin{aligned}\mu'' &= nn' + 2(nm' + n'm) + 4mm', \\ \nu'' &= 2nn' + 4(nm' + n'm) + 4mm', \\ \rho'' &= 4nn' + 4(nm' + n'm) + 2mm', \\ \sigma'' &= 4nn' + 2(nm' + n'm) + mm',\end{aligned}$$

which satisfy the relation $2\mu'' - 2\nu'' + \rho'' - 2\sigma'' = 0$.

And similarly for any number of curves.

43. I consider in particular the case of a cusp, and I denote by $2\kappa 1$ the condition that the conic shall have a cusp (or, what is the same thing, that it shall pass through a given point and touch two given lines through the point). The order and class of the condition $2\kappa 1$ are each $= 3$; hence for the condition $2\kappa 1$ touching a given curve of the order m and class n , we have

$$(\alpha, \beta) = (n + 2m - 3, 2n + 4m - 6).$$

In fact, the number of the conics which have a given cusp and touch a given curve is $= n + 2m$, which, in virtue of the cusp, becomes $n + 2m - 3$; and the number of the conics which have a given cusp and touch a given line is $= 2$, which, in virtue of the cusp, becomes $2 - 2 = 0$ (this belongs to the general theory of characteristics; the number of line-pairs is $2(n + 2m) - (2n + 4m) = 0$, and the number of point-pairs is $2(2n + 4m) - (n + 2m) = 3n + 6m$, which are the line-pairs and point-pairs having a given cusp and touching a given curve).

44. For a system of conics (4Σ) having the characteristics (μ, ν) , the number of cuspidal conics is $= 2\mu - 3\nu$; this may be seen as follows: the characteristics of $2\kappa 1$ are $(3, 3)$, and hence the number of the conics which have a cusp and satisfy the conditions 4Σ is $= 3\mu + 3\nu$; but among these are included the line-pairs, each of which has a cusp at each of the two intersections of the two lines; that is, each line-pair counts as having two cusps; and since the number of line-pairs is $= 2\nu - \mu$, we have therefore

$$3\mu + 3\nu - 2(2\nu - \mu) = 5\mu - \nu,$$

for the number of cuspidal conics, each counted with its proper multiplicity. But the cuspidal conics are each of them a proper conic, and they are therefore counted singly; hence the number is $= 5\mu - \nu$; or, what is the same thing, the number of the conditions $2\kappa 1$ satisfied by the conics of the system (4Σ) is $= 5\mu - \nu$. But I prefer to state the theorem as follows: the number of cuspidal conics in the system (4Σ) is $= 2\mu - 3\nu$; in fact, the number of the conics of the system which pass through a given point is $= \mu$, and among these the number of the line-pairs is $= 2$ (viz. the line-pair is made up of the two lines joining the given point with the two points of the point-pair); hence the number of proper conics through the given point is $= \mu - 2$; and the number of cuspidal conics through the given point is $= (\mu - 2) - 2(\nu - \mu) = 2\mu - 3\nu$.

45. For a system of conics (3Σ) having the characteristics (μ', ν', ρ') , the number of cuspidal conics is $= \mu' - 2\nu' + \frac{1}{2}\rho'$; this may be seen as follows: the characteristics of $(2\kappa 1, \Sigma)$ are $(\mu' - 2\nu' + \frac{1}{2}\rho', \nu' - 2\nu' + \frac{1}{2}\rho', \dots)$, and hence the number of cuspidal conics is obtained as above.

46. In particular, for the system of conics touching three given lines and passing through two given points, the characteristics are $(4, 4)$, and the number of cuspidal conics is $= 2 \cdot 4 - 3 \cdot 4 = -4$, which is $= 0$; the negative value shows that there are no cuspidal conics, which is right, for the conics in question are proper conics.

47. I consider also the case of an inflexion, and I denote by $2\iota 1$ the condition that the conic shall have an inflexion (or, what is the same thing, that it shall touch a given line and pass through two given points on the line). The order and class are each $= 3$; hence for the condition $2\iota 1$ touching a given curve, we have

$$(\alpha, \beta) = (2n + m - 3, 4n + 2m - 6).$$

The formulae for the inflexion are at once obtained from those for the cusp by the interchange of m and n .

48. I give the following formulae, in which $(1^m \kappa^p)$ denotes the system of conditions made up of m contacts 1 and p contacts $2\kappa 1$, and similarly for other symbols; and where the curves are all of them of the order m and class n , and $a = 3n + \kappa$. The expressions for $(1^m \kappa^p)$, &c. are as follows:

$$\begin{aligned} (1) &= n + 2m - 2, \\ (1, 1) &= m^2 + 2mn + \frac{3}{2}n^2 - 3m - \frac{7}{2}n + \frac{1}{2}a, \\ (1, 1, 1) &= 2m^2 + 4mn + n^2 - 8m - 7n + 18 - 3a, \\ (1, 1, 1, 1) &= -4m - 3n - 5 + 3a, \\ (1, 1, 1, 1, 1) &= -8m - 8n - 6 + 6a, \\ (2) &= a, \\ (2, 1) &= 4m + 8n + 44 + a(2m + n - 17), \\ (2, 1, 1) &= 20m + 16n + 42 + a(2m + 2n - 27), \\ (2, 2) &= \frac{1}{2}m^2 + 2m^2n + \frac{7}{2}mn^2 + \frac{5}{2}n^2 - 5m^2 - 9mn - 2n^2 + \frac{9}{2}m + \frac{9}{2}n - 57 + a(-3m - 6n + \frac{9}{2}), \\ (2, 2, 1) &= \frac{1}{2}m^2 + 27m^2n + 2mn^2 + \frac{5}{2}n^2 - 7m^2 - 10mn - 4n^2 - \frac{31}{2}m + \frac{3}{2}n - 54 + a(-3m - 3n + \frac{9}{2}), \\ (3) &= -4m - 5n + 3a, \\ (3, 1) &= -8m - 8n - 6 + 6a, \\ (3, 1, 1) &= -12m - 4n - 12 + 6a, \\ (3, 2) &= -8m^2 - 12mn - 4n^2 + 8m + 16n + 28 + a(4m + 4n - 24), \\ (3, 3) &= -12m^2 - 24mn - 12n^2 + 36m + 60n + 24 + a(6m + 6n - 42), \\ (4) &= 4m + 8n + 44 + a(2m + n - 17), \\ (4, 1) &= 20m + 16n + 42 + a(2m + 2n - 27), \\ (4, 2) &= -8m^2 - 12mn - 4n^2 + 8m + 16n + 28 + a(4m + 4n - 24), \\ (4, 3) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84), \\ (4, 4) &= 8m^2 + 16mn + 8n^2 - 32m - 56n - 40 + a(-4m - 4n + 40), \\ (5) &= -8m - 8n - 6 + 6a, \\ (5, 1) &= -12m - 4n - 12 + 6a, \\ (5, 2) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84), \\ (5, 3) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84), \\ (5, 4) &= 16m^2 + 32mn + 16n^2 - 64m - 112n - 80 + a(-8m - 8n + 80), \\ (5, 5) &= 8m^2 + 16mn + 8n^2 - 32m - 56n - 40 + a(-4m - 4n + 40). \end{aligned}$$

49. I verify that for a conic ($m = n = 2, a = 6$), these values agree with those given by the

general formulae for the conic; thus

$$\begin{aligned}(1) &= 2 + 4 - 2 = 4, \\ (1, 1) &= 4 + 8 + 6 - 6 - 7 + 3 = 8, \\ (1, 1, 1) &= 8 + 16 + 4 - 16 - 14 + 18 - 18 = -2, \text{ \&c.}\end{aligned}$$

50. I do not attempt to verify the expressions for (1^m) by means of the characteristics, but I observe that the formula for $(1, 1, 1, 1, 1)$ may be written

$$-8m - 8n - 6 + 6a = -8m - 8n - 6 + 18n + 6\kappa = -8m + 10n - 6 + 6\kappa,$$

and that for $m = 2, n = 2$, this becomes $-16 + 20 - 6 + 6\kappa = -2 + 6\kappa$, which for $\kappa = 0$ (the proper conic) is $= -2$, agreeing with the value $-8m - 8n - 6 + 6a = -16 - 16 - 6 + 36 = -2$.

51. The expressions for $(1^m \kappa^p)$ are obtained from those for (1^m) by writing therein $m + p$ in place of m , and $n + 2p$ in place of n , and consequently $a + 3p$ in place of a , and then subtracting the number of improper solutions; but the investigation of the improper solutions is in general very difficult.

52. I give the following formulae for the case of a single curve:

$$\begin{aligned}(1\kappa 1) &= n + 2m - 3, \\ (1\kappa 1, 1) &= m^2 + 2mn + \frac{3}{2}n^2 - 5m - \frac{9}{2}n + 6 - \frac{1}{2}a, \\ (1\kappa 1, 1, 1) &= 2m^2 + 4mn + n^2 - 10m - 8n + 24 - 4a, \\ (1\kappa 1, 1, 1, 1) &= -4m - 3n - 8 + 3a, \\ (1\kappa 1, 1, 1, 1, 1) &= -8m - 8n - 18 + 6a, \\ (2\kappa 1) &= a - 7, \\ (2\kappa 1, 1) &= 2a - 8, \\ (2\kappa 1, 1, 1) &= 2a - 4, \\ (2\kappa 1, 2) &= 2m^2 + 2mn + \frac{1}{2}n^2 - 8m - \frac{7}{2}n + 13 - \frac{1}{2}a, \\ (2\kappa 1, 2, 1) &= 2m^2 + 4mn + n^2 - 8m - 7n + 18 - 3a, \\ (2\kappa 1, 2, 2) &= m^2 + 4mn + 2n^2 - 4m - 8n + 12 - 3a, \\ (3\kappa 1) &= -4m - 3n - 5 + 3a, \\ (3\kappa 1, 1) &= -8m - 8n - 6 + 6a, \\ (3\kappa 1, 2) &= -8m^2 - 12mn - 4n^2 + 8m + 16n + 28 + a(4m + 4n - 24), \\ (3\kappa 1, 3) &= -12m^2 - 24mn - 12n^2 + 36m + 60n + 24 + a(6m + 6n - 42), \\ (4\kappa 1) &= 4m + 8n + 44 + a(2m + n - 17), \\ (4\kappa 1, 1) &= 20m + 16n + 42 + a(2m + 2n - 27), \\ (4\kappa 1, 2) &= -8m^2 - 12mn - 4n^2 + 8m + 16n + 28 + a(4m + 4n - 24), \\ (4\kappa 1, 3) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84), \\ (4\kappa 1, 4) &= 8m^2 + 16mn + 8n^2 - 32m - 56n - 40 + a(-4m - 4n + 40), \\ (5\kappa 1) &= -8m - 8n - 6 + 6a, \\ (5\kappa 1, 1) &= -12m - 4n - 12 + 6a, \\ (5\kappa 1, 2) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84), \\ (5\kappa 1, 3) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84), \\ (5\kappa 1, 4) &= 16m^2 + 32mn + 16n^2 - 64m - 112n - 80 + a(-8m - 8n + 80), \\ (5\kappa 1, 5) &= 8m^2 + 16mn + 8n^2 - 32m - 56n - 40 + a(-4m - 4n + 40).\end{aligned}$$

53. I remark that the expressions for $(1^m \kappa^p)$ may be obtained from those for (1^m) by the substitution indicated in No. 51; but the investigation of the improper solutions presents difficulties which I have not attempted to overcome.

54. In the case where the condition Σ is that of passing through a given point, the representatives are $(1, 0)$; and where it is that of touching a given line, the representatives are $(0, 1)$. Hence the characteristics of the system of conics which pass through p given points and touch q given lines ($p + q = 4$) are at once obtained; and the number of cuspidal conics in such a system is $= 2\mu - 3\nu$.

55. I give the following table of the characteristics (μ, ν) for the various systems of conics:

Conditions	μ	ν
(:::)	1	2
(:: /)	2	4
(: //)	4	4
(: ///)	4	2
(// ///)	2	1

56. The characteristics for the systems (1^m) and $(1^m \kappa^p)$ may be obtained by the foregoing formulae; and I have thus obtained the expressions given in the preceding paragraphs.

57. I remark that the whole of the foregoing theory is applicable to curves of any order; but that the formulae for the characteristics are in general very complicated, and have not been obtained except for curves of low order.

ARTICLE NOS. 58 TO 73.

58. I consider the case of a curve of the order r having with a given curve U^m of the order m contacts of given orders; and I denote by $(a, b, c, \dots)$ the number of such curves which have with U^m contacts of the orders $a, b, c, \dots$ respectively, and which besides pass through the requisite number of arbitrary points.

59. De Jonquières has given a formula for the number of curves C^r of the order r which have with a given curve U^m of the order m contacts of the orders $a, b, c, \&c.$ respectively, which besides pass through p points distributed at pleasure on the curve U^m (this includes the case of contacts of any order at given points of the curve U^m), and which moreover satisfy any other $\frac{1}{2}r(r+3) - (a+b+c+\&c.) - p$ conditions; viz. the number of the curves C^r is

$$= \mu \cdot (a+1)(b+1)(c+1) \dots \text{ into}$$

$$\begin{aligned}
& [rm - (a+b+c \dots) - p]^r \\
& + [rm - (a+b+c \dots) - p - 1]^{r-1} (a+b+c \dots) [D]^1 \\
& + [rm - (a+b+c \dots) - p - 2]^{r-2} (ab+ac+bc \dots) [D]^2 \\
& + \dots \\
& + [rm - (a+b+c \dots) - p - t]^0 (abc \dots) [D]^t,
\end{aligned}$$

where the curve U^m is a curve *without cusps*, and having therefore a deficiency $D = \frac{1}{2}(m-1)(m-2)$; the numbers $a, b, c, \dots$ are assumed to be all of them unequal, but if we have α of them each $= a$, β of them each $= b$, $\&c.$, then the foregoing expression is to be divided by $[\alpha]^\alpha [\beta]^\beta \dots$; and μ denotes the number of the curves C^r which satisfy the system of conditions obtained from the given system by replacing the conditions of the contacts of the orders $a, b, c, \&c.$ respectively by the condition of passing through $a+b+c+\dots$ arbitrary points. In order that the formula may

give the number of the proper curves C^r which satisfy the prescribed conditions, it is sufficient that their $\frac{1}{2}r(r+3) - (a+b+c..) - p$ conditions shall include the conditions of passing through at least a certain number T of arbitrary points: this restriction applies to all the formulae of the present section.

60. I will for convenience consider this formula under a somewhat less general form, viz. I will put $p = 0$, and moreover assume that the $\frac{1}{2}r(r+3) - (a+b+c..)$ conditions are the conditions of passing through this number of arbitrary points; whence $\mu = 1$.

We have thus a curve C^r having with the given curve U^m contacts of the orders $a, b, c..$ respectively, and besides passing through $\frac{1}{2}r(r+3) - (a+b+c..)$ arbitrary points; and the number of such curves is by the formula $= (a+1)(b+1)(c+1) \dots$ into

$$\begin{aligned} & [rm - (a+b+c..)]^r \\ & + [rm - (a+b+c..) - 1]^{r-1}(a+b+c..)[D]^1 \\ & + [rm - (a+b+c..) - 2]^{r-2}(ab+ac+bc..)[D]^2 \\ & + \dots \\ & + [rm - (a+b+c..) - t]^0(abc..)[D]^t, \end{aligned}$$

where, as before, in the case of any equalities between the numbers $a, b, c, \dots$, the expression is to be divided by $[\alpha]^\alpha[\beta]^\beta \dots$

61. I have succeeded in extending the formula to the case of a curve with cusps; instead of writing down the general formula, I will take successively the cases of a single contact a , two contacts a, b , three contacts a, b, c , &c.; and then denoting the numbers of the curves by (a) , (a, b) , (a, b, c) , &c. in these cases respectively, I say that we have

$$(a) = (a+1)[rm - a]^r[1 + aD]^r - a\kappa,$$

$$\begin{aligned} (a, b) = (a+1)(b+1) \{ & [rm - a - b]^r \\ & + [rm - a - b - 1]^{r-1}(a+b)[D]^1 \\ & + ab[D]^2 \} \\ & - a(b+1) \{ [rm - a - b - 1]^{r-1}[1 + bD]^1 \} \\ & - b(a+1) \{ [rm - a - b - 1]^{r-1}[1 + aD]^1 \} \\ & + ab\kappa', \end{aligned}$$

$$\begin{aligned}
(a, b, c) = & (a+1)(b+1)(c+1) \left\{ [rm - a - b - c]^r \right. \\
& + [rm - a - b - c - 1]^{r-1} (a+b+c) [D]^1 \\
& + [rm - a - b - c - 2]^{r-2} (ab+ac+bc) [D]^2 \\
& \quad \left. + abc [D]^3 \right\} \\
& - c(a+1)(b+1) \left\{ [rm - a - b - c - 1]^{r-1} \right. \\
& \quad + [rm - a - b - c - 2]^{r-2} (a+b) [D]^1 \\
& \quad \left. + ab [D]^2 \right\} \\
& - [2bc(a+1)] \left\{ [rm - a - b - c - 2]^{r-2} \right. \\
& \quad + [rm - a - b - c - 3]^{r-3} a [D]^1 \left. \right\} [\kappa]^1 \\
& - [2ac(b+1)] \left\{ [rm - a - b - c - 2]^{r-2} \right. \\
& \quad + [rm - a - b - c - 3]^{r-3} b [D]^1 \left. \right\} [\kappa]^1 \\
& - [2ab(c+1)] \left\{ [rm - a - b - c - 2]^{r-2} \right. \\
& \quad + [rm - a - b - c - 3]^{r-3} c [D]^1 \left. \right\} [\kappa]^1 \\
& \quad \quad \quad + abck'.
\end{aligned}$$

77. The foregoing examples are sufficient to exhibit the law; but as I shall have to consider the cases of four and five contacts, I will also write down the formula for (a, b, c, d) , putting therein for shortness

$$\begin{aligned}
a+b+c+d &= \alpha, & ab+\dots+cd &= \beta, & abc+\dots+bcd &= \gamma, & abcd &= \delta, \\
a+b+c &= \alpha', & ab+ac+bc &= \beta', & abc &= \gamma', & a+b &= \alpha'', & ab &= \beta'', & a &= \alpha''';
\end{aligned}$$

and also the formula for (a, b, c, d, e) , putting therein in like manner

$$(\alpha, \beta, \gamma, \delta, \epsilon), (\alpha', \beta', \gamma', \delta'), (\alpha'', \beta'', \gamma''), (\alpha''', \beta'''), (\alpha''')$$

for the combinations of (a, b, c, d, e) , (a, b, c, d) , (a, b, c) , (a, b) and (a) respectively.

We have

$$\begin{aligned}
(a, b, c, d) = & (a+1)(b+1)(c+1)(d+1) \left\{ [rm - \alpha]^r \right. \\
& + [rm - \alpha - 1]^{r-1} \alpha [D]^1 \\
& + [rm - \alpha - 2]^{r-2} \beta [D]^2 \\
& + [rm - \alpha - 3]^{r-3} \gamma [D]^3 \\
& \quad \quad \quad + \delta [D]^4 \left. \right\} \\
& - [2d(a+1)(b+1)(c+1)] \left\{ [rm - \alpha - 1]^{r-1} \right. \\
& \quad + [rm - \alpha - 2]^{r-2} \alpha' [D]^1 \\
& \quad + [rm - \alpha - 3]^{r-3} \beta' [D]^2 \\
& \quad \quad \quad + \gamma' [D]^3 \left. \right\} [\kappa]^1 \\
& + [2cd(a+1)(b+1)] \left\{ [rm - \alpha - 2]^{r-2} \right. \\
& \quad + [rm - \alpha - 3]^{r-3} \alpha'' [D]^1 \\
& \quad \quad \quad + \beta'' [D]^2 \left. \right\} [\kappa]^2
\end{aligned}$$

$$\begin{aligned}
& - [2bcd(a+1)] \{ [rm - \alpha - 3]^{r-3} \\
& \qquad \qquad \qquad + \alpha''' [D]^1 \} [\kappa]^3 \\
& \qquad \qquad \qquad + abcd\kappa' [\kappa]^4.
\end{aligned}$$

78. In all these formulae there is, as before, a numerical divisor in the case of any equalities among the numbers $a, b, c, \&c.$ And D denotes, as before, the deficiency, viz. its value now is $D = \frac{1}{2}(m-1)(m-2) - \delta - \kappa$; or observing that the class n is $= m^2 - m - 2\delta - 3\kappa$, we have $D = \frac{1}{2}n - m + 1 + \frac{1}{2}\kappa$, or say $D = 1 + A$ if $A = -m + \frac{1}{2}n + \frac{1}{2}\kappa$.

79. It is to be observed with reference to the applicability of these formulae within certain limits only, that the formulae are the *only* formulae which are generally true; thus taking the simplest case, that of a single contact a , the only algebraical expression for the number of the curves C^r which have with a given curve U^m a contact of the order a , and besides pass through the requisite number $\frac{1}{2}r(r+3) - a$ of arbitrary points, is that given by the formula, viz.

$$(a) = (a+1)(rm - a + aD) - a\kappa.$$

Considering the curve U^m and the order r of the curve C^r as given, if a has successively the values $1, 2, \dots$ up to a limiting value of a , the formula gives the number of the proper curves C^r which have with the given curve U^m a contact of the required order a : beyond this limiting value the formula no longer gives the number of the proper curves C^r which satisfy the required condition, and it thus ceases to be applicable; but there is no algebraic function of a which would give the number of the proper curves C^r as well beyond as up to the foregoing limiting value of a .

80. The formulae are applicable provided only the conditions include the conditions of passing through a sufficient number of arbitrary points; viz. when the number of arbitrary points is sufficiently great, it is not possible to satisfy the conditions specially by means of improper curves C^r , being or comprising a pair of coincident curves. Thus to take a simple example, suppose it is required to find the number of the conics which touch a given curve m times and besides pass through $\sigma - m$ given points: if the number of the given points be 4 or 3 there is no coincident line-pair through the given points, and therefore no coincident line-pair satisfying the given conditions; if the number of the given points is $= 2$, then the line joining these points gives a coincident line-pair having at each of its m intersections with the given curve a special contact therewith, that is, having in $\frac{1}{6}m(m-1)(m-2)$ ways three special contacts with the given curve; if the number of the given points is 1 or 0, then in the first case any line whatever through the given point, and in the second case any line whatever, regarded as a coincident line-pair, has m special contacts with the given curve; and so in general there is a certain value for the number of given points, for which value the conditions of contact may be satisfied by a determinate number of improper curves C^r , and for values inferior to it the conditions may be satisfied by infinite series of improper curves C^r . It is by such considerations as these that De Jonquières has determined the minimum value T of the number of arbitrary points to which the conditions should relate in order that the formulae may be applicable: I refer for his investigation and results to paragraphs XVII and XVIII of his memoir.

I remark that in the case where the number of improper solutions is finite, the formula can be corrected so as to give the number of proper solutions by simply subtracting the number of the improper solutions: but this is not so when the improper solutions are infinite in number; the mode of obtaining the approximate formulae in this case is very complicated.

81. I have mentioned that the formulae are applicable within certain limits only; and I will now explain how the limits may be determined in the case of a single contact a . Writing for shortness $A = -m + \frac{1}{2}n + \frac{1}{2}\kappa$, that is, $D = 1 + A$, we have

$$(a) = (a+1)(rm - a + a + aA) - a\kappa = (a+1)(rm) - (a+1)a + (a+1)aA - a\kappa.$$

The expression $(a+1)(rm) - (a+1)a$ is always positive for values of a from $a = 0$ up to a certain limiting value; but for values of a beyond this, the expression becomes negative, and the number of proper curves is $= 0$. The condition that the formula shall give a positive result is

$$(a+1)(rm - a + aA) > a\kappa,$$

or, as this may be written,

$$(a+1)(rm) - (a+1)a + a[(a+1)A - \kappa] > 0.$$

82. In particular, for a conic ($r = 2$) and a proper curve without singularities ($\kappa = 0$, $\delta = 0$, $n = m(m-1)$, $A = -m + \frac{1}{2}m(m-1) = \frac{1}{2}(m^2 - 3m)$), the formula becomes

$$(a) = (a+1)(2m - a) + (a+1)a \cdot \frac{1}{2}(m^2 - 3m),$$

and the condition of a positive result is

$$(a+1)(2m - a) + \frac{1}{2}(a+1)a(m^2 - 3m) > 0,$$

which is always satisfied for positive values of a ; that is, for a conic and a proper curve, the formula is applicable for all values of a .

83. For a conic and a curve with cusps, we have $\kappa > 0$, and the formula may cease to be applicable for large values of a ; thus for a cuspidal cubic ($m = 3$, $n = 6$, $\kappa = 1$, $\delta = 0$, $A = -3 + 3 + \frac{1}{2} = \frac{1}{2}$), we have

$$(a) = (a+1)(6 - a + \frac{1}{2}a) - a = (a+1)(6 - \frac{1}{2}a) - a = 6 + 5a - \frac{1}{2}a^2 - \frac{1}{2}a = 6 + \frac{9}{2}a - \frac{1}{2}a^2,$$

which is positive for $a = 1, 2, \dots, 9$, but negative for $a = 10$; that is, the formula ceases to be applicable for $a = 10$.

84. I remark that the formulae for (a) , (a, b) , &c. may be written in a different form; viz. if we introduce the new symbols $[a]$, $[a, b]$, &c. defined by the equations

$$\begin{aligned} (a) &= [a], \\ (a, b) &= [a][b] + [a, b], \\ (a, b, c) &= [a][b][c] + [a, b][c] + [a, c][b] + [b, c][a] + [a, b, c], \\ &\text{\&c.}, \end{aligned}$$

then the expressions for $[a]$, $[a, b]$, $[a, b, c]$, &c. are much simpler than those for (a) , (a, b) , (a, b, c) , &c.

85. I do not attempt to write down the general expressions for $[a]$, $[a, b]$, &c., but I remark that they may be obtained by elimination from the foregoing equations; and that they are of great importance in the theory.

86. I give the following formulae for the characteristics of the systems of conics which satisfy given conditions of contact with a given curve; the characteristics are expressed in terms of m , n , and $a = 3n + \kappa$:

$$\begin{aligned} \mu(1) &= n + 2m - 2, & \nu(1) &= 2n + 4m - 4, \\ \mu(1, 1) &= m^2 + 2mn + \frac{3}{2}n^2 - 3m - \frac{7}{2}n + \frac{1}{2}a, & \nu(1, 1) &= 2m^2 + 4mn + n^2 - 6m - 5n + a, \\ \mu(2) &= a, & \nu(2) &= 2a, \\ \mu(2, 1) &= 4m + 8n + 44 + a(2m + n - 17), & \nu(2, 1) &= 12m + 24n + 96 + a(4m + 2n - 36), \\ \mu(3) &= -4m - 5n + 3a, & \nu(3) &= -12m - 15n + 9a. \end{aligned}$$

87. The characteristics for the higher systems may be obtained by the formulae of Nos. 38, 39; but the calculations are very complicated.

88. I give the following formulae for the characteristics of the systems of conics which have a cusp and satisfy given conditions:

$$\begin{aligned}\mu(1\kappa 1) &= n + 2m - 3, & \nu(1\kappa 1) &= 2n + 4m - 6, \\ \mu(1\kappa 1, 1) &= m^2 + 2mn + \frac{3}{2}n^2 - 5m - \frac{9}{2}n + 6 - \frac{1}{2}a, & \nu(1\kappa 1, 1) &= 2m^2 + 4mn + n^2 - 10m - 8n + 18 - 2a, \\ \mu(2\kappa 1) &= a - 7, & \nu(2\kappa 1) &= 2a - 14.\end{aligned}$$

89. The characteristics for the higher systems may be obtained as before; but the calculations are even more complicated than in the case of proper conics.

90. I remark that the formulae for the characteristics are applicable only within certain limits; and that beyond these limits the formulae require correction. The determination of these corrections is a problem of great difficulty, which I have not attempted to solve in general.

91. I give the following table of the values of (1^m) and $(1^m\kappa^p)$ for a conic and a given curve; the values are expressed in terms of m , n , and $a = 3n + \kappa$:

$$\begin{aligned}(1) &= n + 2m - 2, \\ (1, 1) &= m^2 + 2mn + \frac{3}{2}n^2 - 3m - \frac{7}{2}n + \frac{1}{2}a, \\ (1, 1, 1) &= 2m^2 + 4mn + n^2 - 8m - 7n + 18 - 3a, \\ (1, 1, 1, 1) &= -4m - 3n - 5 + 3a, \\ (1, 1, 1, 1, 1) &= -8m - 8n - 6 + 6a, \\ (2) &= a, \\ (2, 1) &= 4m + 8n + 44 + a(2m + n - 17), \\ (2, 1, 1) &= 20m + 16n + 42 + a(2m + 2n - 27), \\ (3) &= -4m - 5n + 3a, \\ (3, 1) &= -8m - 8n - 6 + 6a, \\ (3, 1, 1) &= -12m - 4n - 12 + 6a, \\ (4) &= 4m + 8n + 44 + a(2m + n - 17), \\ (4, 1) &= 20m + 16n + 42 + a(2m + 2n - 27), \\ (5) &= -8m - 8n - 6 + 6a.\end{aligned}$$

92. The question which ought now to be considered is to determine the corrections or supplements which should be applied to the foregoing expressions (a) , (a, b) , &c., or to their equivalents $[a]$, $[a][b] + [a, b]$, &c. in order to obtain formulae for the cases beyond the limits within which the present formulae are applicable; but this I am not in a position to enter upon. If the extended formulae were obtained, it would of course be an interesting verification or application of them to deduce from them the complete series of expressions (1) , (2) , $\dots$, $(1, 1, 1, 1, 1)$ for the number of the conics which satisfy given conditions of contact with a given curve, and besides pass through the requisite number of given points. It will be recollected that throughout these last investigations, I have put De Jonquières' $p = 0$; that is, I have not considered the case of the curves C^r which (among the conditions satisfied by them) have with the curve U^m contacts of given orders at given points of the curve; it is probable that the general formulae containing the number p admit of extensions and transformations analogous to the formulae in which p is put $= 0$, but this is a question which I have not considered.

93. The set of equations $(a) = [a]$, $(a, b) = [a][b] + [a, b]$, &c., considered irrespectively of the meaning of the symbols contained therein, gives rise to an analytical question which is considered in Annex No. 7.

The question of the conics satisfying given conditions of contact is considered from a different point of view in my Second Memoir above referred to.

Annex No. 1 (referred to in the notice of De Jonquières' memoir of 1861).—**On the form of the equation of the curves of a series of given index.**

To obtain the general form of the equation of the curves C^n of a series of the index N , it is to be observed that the equation of any such curve is always included in an equation of the order n in the coordinates, containing linearly and homogeneously certain parameters $a, b, c, \dots$; this is universally the case, as we may, if we please, take the parameters $(a, b, c, \dots)$ to be the coefficients of the general equation of the order n ; but it is convenient to make use of any linear relations between these coefficients so as to reduce as far as possible the number of the parameters. Assume that the number of the parameters is $= \omega + 1$, then in order that the curve should form a series (that is, satisfy $\frac{1}{2}n(n+3) - 1$ conditions), we must have a $(\omega - 1)$ -fold relation between the parameters, or, what is the same thing, taking the parameters to be the coordinates of a point in ω -dimensional space, say the parametric point, the point in question must be situate on a $(\omega - 1)$ -fold locus. Moreover, the condition that the curve shall pass through a given point establishes between the parameters a linear relation (viz. that expressed by the original equation of the curve regarding the coordinates therein as belonging to the given point, and therefore as constants); that is, when the curve passes through a given point, the corresponding positions of the parametric point are given as the intersections of the $(\omega - 1)$ -fold locus by an $(\omega - 1)$ -fold locus; the number of the curves is therefore equal to the number of these intersections, that is, to the order of the $(\omega - 1)$ -fold locus; or the index of the series being assumed to be $= N$, the order of the $(\omega - 1)$ -fold locus must be also $= N$. That is, the general form of the equation of the curves C^n which form a series of the index N , is that of an equation of the order n containing linearly and homogeneously the $\omega + 1$ coordinates of a certain $(\omega - 1)$ -fold locus of the order N . It is only in a particular case, viz. that in which the $(\omega - 1)$ -fold locus is unicursal, that the coordinates of a point of this locus can be expressed as rational and integral functions of the order N of a variable parameter θ ; and consequently only in this same case that the equation of the curves C^n of the series of the index N can be expressed by an equation $(* \overset{\omega}{\curvearrowright} x, y, z)^n = 0$, or $(* \overset{\omega}{\curvearrowright} x, y, 1)^n = 0$, rational and integral of the degree N in regard to a variable parameter θ .

If in the general case we regard the coordinates of the parametric point as irrational functions of a variable parameter θ , then rationalising in regard to θ , we obtain an equation rational of the order N in θ , but the order in the coordinates instead of being $= n$, is equal to a multiple of n , say qn . Such an equation represents not a single curve but q distinct curves C^n , and it is to be observed that if we determine the parameter by substituting therein for the coordinates their values at a given point, then to each of the N values of the parameter there corresponds a system of q curves, only one of which passes through the given point, the other $q - 1$ curves are curves not passing through the given point, and having no proper connexion with the curves which satisfy this condition.

Returning to the proper representation of the series by means of an equation containing the coordinates of the parametric point, say an equation $(* \overset{\omega}{\curvearrowright} x, y, 1)^n = 0$, involving the two coordinates (x, y) , it is to be noticed that forming the derived equation and eliminating the coordinates of the parametric point, we obtain an equation rational in the coordinates (x, y) , and also rational of the degree N in the differential coefficient $\frac{dy}{dx}$; in fact since the number of curves through any given point (x_0, y_0) is $= N$, the differential equation must give this number of directions of passage from the point (x_0, y_0) to a consecutive point, that is, it must give this number of values of $\frac{dy}{dx}$, and must consequently be of the order N in this quantity.

Conversely, if a given differential equation rational in $x, y, \frac{dy}{dx}$, and of the degree N in the last-mentioned quantity $\frac{dy}{dx}$, admit of an algebraical general integral, the curves represented by this integral equation may be taken to be irreducible curves, and this being so they will be curves of a certain order n forming a series of the index N ; whence the general integral (assumed to be algebraical) is given by an equation of the above-mentioned form, viz. an equation rational of a certain order n in the coordinates, and containing linearly and homogeneously the $\omega + 1$ coordinates of a variable parametric point situate on an $(\omega - 1)$ -fold locus. The integral equation expressed in the more usual form of an equation rational of the order N in regard to the parameter or constant of integration, will be in regard to the coordinates of an order equal to a multiple of n , say $= qn$, and for any given value of the parameter will represent not a single curve C^n , but a system of q such curves: the first-mentioned form is, it is clear, the one to be preferred.

Annex No. 2 (referred to, No. 17).—**On the line-pairs which pass through three given points and touch a given conic.**

Taking the given points to be the angles of the triangle formed by the lines $(x = 0, y = 0, z = 0)$, we have to find (f, g, h) such that the conic $(0, 0, 0, f, g, h \mid x, y, z)^2 = 0$, or, what is the same thing, $xyz + gzx + hxy = 0$, shall reduce itself to a line-pair, and shall touch a given conic $(1, 1, 1, \lambda, \mu, \nu \mid x, y, z)^2 = 0$. The condition for a line-pair is that one of the quantities f, g, h shall vanish, viz. it is $fgh = 0$; the condition for the contact of the two conics is found in the usual manner by equating to zero the discriminant of the function

$$\lambda(1 + \theta f) \cdot \mu(1 + \theta g) \cdot \nu(1 + \theta h) - \dots = (a, b, c, d \mid e, \dots)$$

suppose; the values of a, b, c, d being

$$\begin{aligned} a &= -(f^2 + g^2 + h^2 - 2\lambda gh - 2\mu hf - 2\nu fg), \\ b &= ((\mu\nu - \lambda)f + (\nu\lambda - \mu)g + (\lambda\mu - \nu)h), \\ c &= 1 - \lambda^2 - \mu^2 - \nu^2 + 2\lambda\mu\nu, \\ d &= 2fgh. \end{aligned}$$

Hence considering (f, g, h) as the coordinates of the parametric point, we have the discriminant-locus $a = 0$, and the contact-locus

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd = 0,$$

and at the intersection of the two loci, $a = 0$, $b^2(4bd - 3c^2) = 0$, equations breaking up into the system $(a = 0, b = 0)$ twice, and the system $a = 0$, $4bd - 3c^2 = 0$; the former of these is

$$fgh = 0, \quad f^2 + g^2 + h^2 - 2\lambda gh - 2\mu hf - 2\nu fg = 0,$$

which expresses that the intersection of the two lines of the line-pair intersect on the given conic; in fact the system is satisfied by $f = 0$, $g^2 + h^2 - 2\lambda gh = 0$, giving a line-pair $x(hy + gz) = 0$, the two lines whereof intersect on the conic $(1, 1, 1, \lambda, \mu, \nu \mid x, y, z)^2 = 0$; and similarly, if $g = 0$, then $h^2 + f^2 - 2\mu hf = 0$, or if $h = 0$, then $f^2 + g^2 - 2\lambda fg = 0$. As noticed above this system occurs twice.

The second system $a = 0$, $4bd - 3c^2 = 0$ gives

$$\begin{aligned} fgh = 0, \quad -2fgh((\mu\nu - \lambda)f + (\nu\lambda - \mu)g + (\lambda\mu - \nu)h) \\ - 3(1 - \lambda^2 - \mu^2 - \nu^2 + 2\lambda\mu\nu)^2 = 0, \end{aligned}$$

or, as the second equation may also be written,

$$f^2 = (1 - \mu^2)(1 - \nu^2) + g^2(\lambda^2 - 1)(1 - \nu^2) + h^2(1 - \lambda^2)(1 - \mu^2) \\ + 2gh(1 - \lambda^2)(\mu\nu - \lambda) + 2hf(1 - \mu^2)(\nu\lambda - \mu) + 2fg(1 - \nu^2)(\lambda\mu - \nu) = 0,$$

which expresses that a line of the line-pair touches the conic; in fact the system is satisfied by $f = 0$, $g^2(1 - \nu^2) + h^2(1 - \mu^2) + 2gh(\mu\nu - \lambda) = 0$, viz. we have here the line-pair $x(hy + gz) = 0$, in which the line $hy + gz = 0$ touches the conic $(1, 1, 1, \lambda, \mu, \nu \mid x, y, z)^2 = 0$ and the like if $g = 0$, or if $h = 0$. This system it has been seen occurs only once.

Annex No. 3 (referred to, No. 22).—**On the conics which pass through two given points and touch a given conic.**

Consider the conics which pass through two given points and touch a given conic. We may take $Z = 0$ as the equation of the line through the two given points, and then taking the pole of this line in regard to the given conic and joining it with the two given points respectively, the equations of the joining lines may be taken to be $X = 0$ and $Y = 0$ respectively. This being so, we have for the given points $(X = 0, Z = 0)$ and $(Y = 0, Z = 0)$ respectively, and for the given conic

$$aX^2 + bY^2 + 2hXY + cZ^2 = 0;$$

and since the required conic is to pass through the two given points its equation will be of the form

$$wX^2 + 2xYZ + 2yZX + 2zXY = 0,$$

where (x, y, z, w) are variable parameters which must satisfy a single condition in order that the last-mentioned conic may touch the given conic. The condition is at once seen to be that obtained by making the equation

$$(a + \lambda w)(b + \lambda w) - (h + \lambda z)^2 - c(x + \lambda y)^2 = 0,$$

considered as a cubic equation in λ , have a pair of equal roots; or if we write

$$A = 3(ab - h^2), \\ B = (ab - h^2)w - 2chz, \\ C = -ax^2 - by^2 - cz^2 + 2h(xy - zw), \\ D = 3z(2xy - wz),$$

then the required condition is

$$A^2D^2 + 4AC^3 + 4B^3D - 3B^2C^2 - 6ABCD = 0.$$

Hence the conic

$$wX^2 + 2xYZ + 2yZX + 2zXY = 0$$

satisfies the prescribed conditions, if only the parameters (x, y, z, w) satisfy the last-mentioned equation, that is, if (x, y, z, w) are the coordinates of a point on the sextic surface represented by this equation.

The surface has upon it a cuspidal curve the equations whereof are $A = 0$, $B = 0$, $C = 0$, $D = 0$; this may be considered as the intersection of the quadric surface $AC - B^2 = 0$ and the cubic surface $AD - BC = 0$; and the cuspidal curve is consequently a sextic.

The surface has also a nodal curve made up of two conics; to prove this I write for shortness $k = h - \sqrt{ab}$, $k_1 = h + \sqrt{ab}$; the values of A, B, C, D then are

$$\begin{aligned} A &= -kk_1, \\ B &= -kk_1w - 2chz, \\ C &= -ax^2 - by^2 - cz^2 + 2h(xy - zw), \\ D &= 3z(2xy - zw); \end{aligned}$$

and it is in the first place to be shown that the surface contains the conic

$$(x : y : z : w) = \theta\sqrt{b} : \theta\sqrt{a} : 1 : -k\theta^2 + t,$$

where θ is a variable parameter. Substituting these values, we have

$$\begin{aligned} B &= k_1(k\theta^2 - c(\sqrt{a} + \sqrt{ab})), \\ C &= 2kk_1\theta^2 - (3h - \sqrt{ab}), \end{aligned}$$

and hence

$$\begin{aligned} AD - BC &= -2k(kk_1\theta^2 + 2c\sqrt{ab})^2, \\ AC - B^2 &= -k^2(kk_1\theta^2 + 2c\sqrt{ab})^2, \\ BD - C^2 &= (kk_1\theta^2 + 2c\sqrt{ab})^2, \end{aligned}$$

values which satisfy identically the equation of the surface written under the form $(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0$.

Moreover, proceeding to form the derived equation, and to substitute therein the foregoing values of (x, y, z, w) , we have

$$dA : dB : dC : dD = k^2 : 2k : 1 : 0,$$

and then the derived equation is

$$\begin{aligned} (AD - BC)(3A - 2kB - k^2C) - 2(AC - B^2)(3B - 4kC + k^2D) \\ - 2(BD - C^2)(2kA - 2k^2B) = 0, \end{aligned}$$

that is,

$$-k(3A - 2kB - k^2C) + (2kA - 2k^2B) = 0,$$

or finally

$$-k(A - 3Bk + 3Ck^2 - Dk^3) = 0,$$

which is satisfied by the foregoing values of A, B, C, D ; hence the conic is a nodal curve on the sextic; and by merely changing the sign of one of the radicals $\sqrt{a}$, $\sqrt{b}$ (and therefore interchanging k, k_1) we obtain another conic which is also nodal curve on the surface, that is, we have as nodal curves the two conics

$$\begin{aligned} (x : y : z : w) &= \theta\sqrt{b} : \theta\sqrt{a} : 1 : -k\theta^2 + t, \\ (x : y : z : w) &= \theta\sqrt{b} : -\theta\sqrt{a} : 1 : -k_1\theta^2 + t. \end{aligned}$$

It is to be remarked that each of the nodal conics meets the cuspidal curve in two points, viz. writing for shortness $\theta = \frac{1}{2}\sqrt{\frac{c}{h}}$, $\theta_1 = -\frac{1}{2}\sqrt{\frac{c}{h}}$, for the intersections of the first conic we have

$$(x : y : z : w) = \theta\sqrt{a} : \theta\sqrt{b} : 1 : t \quad \text{and} \quad = -\theta\sqrt{a} : -\theta\sqrt{b} : 1 : t,$$

and for the intersections with the second conic

$$(x : y : z : w) = \theta_1 \sqrt{a} : -\theta_1 \sqrt{b} : 1 : t \quad \text{and} \quad = -\theta_1 \sqrt{a} : \theta_1 \sqrt{b} : 1 : t.$$

The condition of passing through any arbitrary point establishes a linear relation between the parameters (x, y, z, w) . Hence, if the conic in addition to the prescribed conditions passes through two other given points, the point (x, y, z, w) is given as the intersection of a line with the sextic surface; the number of intersections is $= 6$. If (x, y, z, w) is situate on the cuspidal curve, then the conic instead of simply touching the given conic will have with it a contact of the second order, and if we besides suppose that the conic passes through a given point, then the point (x, y, z, w) is given as the intersection of the cuspidal curve with a plane; the number is $= 6$. Similarly, if the conic has two contacts with the given conic, and besides passes through a given point, then the point (x, y, z, w) is given as the intersection of the nodal curve by a plane; the number is $= 4$. Finally (observing that in the case in question of the contacts of a conic with a conic we cannot have three simple contacts, or a simple contact and one of the second order), a point of intersection of the nodal and cuspidal curves answers to a contact of the third order; and the number is $= 4$. That is, the theory

of the sextic surface leads to the following values (agreeing with those obtained from the formulae by writing therein $m = n = 2$, $a = 6$), viz.

$$\begin{aligned} (1 ::) &= 6, = 2m + n, \\ (1, 1, \cdot) &= 4, = 2m^2 + 2mn + \frac{1}{2}n^2 - 2m - \frac{5}{2}n + a, \\ (2, \cdot) &= 6, = a, \\ (3 :) &= 4, = -4m - 5n + 3a. \end{aligned}$$

I remark that the section by an arbitrary plane is a sextic curve having 6 cusps and 4 nodes; it is therefore a *unicursal* sextic; this suggests the theorem that the sextic surface is also unicursal, viz. that the coordinates are expressible rationally in terms of two parameters; I have found that this is in fact the case. In doing this there is no loss of generality in supposing that $a = b = c = 1$; and assuming that this is so, and putting also $-1 + h = k$, $1 + h = k_1$, and therefore $2h = k + k_1$, we have

$$\begin{aligned} A &= -kk_1, \\ B &= -kk_1w - (k + k_1)z, \\ C &= -x^2 - y^2 - z^2 + (k + k_1)(xy - zw), \\ D &= 3z(2xy - zw). \end{aligned}$$

The equation of the sextic surface being, as before,

$$A^2D^2 + 4AC^3 + 4B^3D - 3B^2C^2 - 6ABCD = 0,$$

I say that this equation is satisfied on writing therein

$$\begin{aligned} x + y &= \sqrt{-k}(1 + k_1a) \sin \phi, \\ x - y &= \sqrt{k}(1 + ka) \cos \phi, \\ z &= 1, \\ w &= (2a - 1) \cos^2 \phi + (2a + 1) \sin^2 \phi, \end{aligned}$$

where (a, ϕ) are arbitrary. In fact these values give

$$\begin{aligned} A &= -kk_1 \cos^2 \phi - kk_1 \sin^2 \phi, \\ B &= -k(2k_1a + 1) \cos^2 \phi - k_1(2ka + 1) \sin^2 \phi, \\ C &= -k(ak_1 + 2) \cos^2 \phi - k_1(ak + 2) \sin^2 \phi, \\ D &= -ka^2 \cos^2 \phi - k_1a^2 \sin^2 \phi, \end{aligned}$$

whence, ω being arbitrary, we have

$$(A, B, C, D \mid \omega, 1)^3 = -[k \cos^2 \phi(k_1 \omega + 1) + k_1 \sin^2 \phi(k \omega + 1)](\omega + a)^2,$$

viz. the equation $(A, B, C, D \mid \omega, 1)^3 = 0$, considered as a cubic equation in ω , has the twofold root $\omega = -a$, that is, we have the above relation between (A, B, C, D) . Whence also writing $\sin \phi = \frac{2\lambda}{1+\lambda^2}$, $\cos \phi = \frac{1-\lambda^2}{1+\lambda^2}$, the equation of the surface is satisfied by the values

$$\begin{aligned} x + y : x - y : z : w &= \sqrt{-k}(1 + k_1 a) \cdot 2\lambda(1 + \lambda^2) : \sqrt{k}(1 + ka)(1 - \lambda^2)(1 + \lambda^2) \\ &: (1 + \lambda^2)^2 : (1 + \lambda^2)^2[(2a - 1) \cos^2 \phi + (2a - 1) \sin^2 \phi], \end{aligned}$$

or the coordinates are expressed rationally in terms of a, λ .

Annex No. 4 (referred to, Nos. 22 and 71).—**On the Conics which touch a cuspidal cubic.**

In the cuspidal cubic, if $x = 0$ be the equation of the tangent at the cusp, $y = 0$ that of the line joining the cusp with the inflexion, and $z = 0$ that of the tangent at the cusp, then the equation of the curve is $y^2 = x^2 z$; the coordinates of a point on the cubic are given by $x : y : z = 1 : \theta : \theta^3$, where θ is a variable parameter; and we have, at the cusp $\theta = \infty$, at the inflexion $\theta = 0$. In the cubic, $m = n = 3$, $a(= 3n + \kappa) = 10$.

Considering now the conic

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

this meets the cubic in the 6 points the parameters of which are determined by the equation

$$(a, b, c, f, g, h \mid 1, \theta, \theta^3)^2 = 0,$$

or, what is the same thing,

$$(c, f, g, b, 2h, a \mid \theta, 1)^6 = 0.$$

The discriminant of this sextic function contains the factor c , hence equating the residual factor to zero, we obtain the equation of the contact-locus in the form

$$(c, f, g, b, h, a)^6 = 0.$$

It follows that the number of the conics $(1 ::)$ is $= 9$, which agrees with the general value $(1 ::) = 2m + n$. If the conic pass through the cusp we have $c = 0$, and the equation in θ is reduced to a quartic; it is convenient to alter the letters in such wise that the quartic equation may be obtained in the standard form $(a, b, c, d, e \mid \theta, 1)^4 = 0$; viz. this will be the case if the equation of the conic is taken to be

$$(e, 6c, 0, 6d, 2b, 0 \mid x, y, z)^2 = 0,$$

and we then obtain the equation of the contact-locus in the form

$$(ae - 4bd + 3c^2)^3 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0,$$

which is a one-fold locus of the order 6. It follows that we have

$$(1\kappa 1, 1, \cdot) = 6, \quad \text{agreeing with } (1\kappa 1, 1, \cdot) = n + 2m - 3.$$

The condition in order that the conic may touch a given line is given by an equation of the form

$$(* \mid a^2, ab, b^2, 2ce - 3d^2, ae - 8bd, ad - 12bc)^2 = 0,$$

which is a one-fold locus of the order 2; it at once follows that we have

$$(1\kappa 1, 1, /) = 12, \quad \text{agreeing with } (1\kappa 1, 1, /) = 2n + 4m - 6.$$

It is a matter of some difficulty to show that we have

$$(1\kappa 1, 1, //) = 18, \quad \text{agreeing with } (1\kappa 1, 1, //) = 4m + 4n - 6;$$

but I proceed to effect this, first remarking that I do not attempt to prove the remaining case

$$(1\kappa 1, 1, ///) = 15, \quad \text{agreeing with } (1\kappa 1, 1, ///) = 4m + 2n - 3.$$

Investigation of the value $(1\kappa 1, 1, //) = 18$:

We have the sextic locus

$$(ae - 4bd + 3c^2)^3 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0,$$

and combined therewith two quadric loci,

$$\begin{aligned} (* \mid a^2, ab, b^2, 2ce - 3d^2, ae - 8bd, ad - 12bc)^2 &= 0, \\ (*' \mid a^2, ab, b^2, 2ce - 3d^2, ae - 8bd, ad - 12bc)^2 &= 0, \end{aligned}$$

which intersect in a three-fold locus of the order 24; it is to be shown that this contains as part of itself the quadric three-fold locus $(a = 0, b = 0, 2ce - 3d^2 = 0)$ taken three times, leaving a residual locus of the order $24 - 6 = 18$.

We may imagine the coordinates a, b, c, d, e expressed as linear functions of any four coordinates, and so reduce the problem from a problem in 4-dimensional space to one in ordinary 3-dimensional space. We have thus a sextic surface, and two quadric surfaces; the sextic is a developable surface or torse, having for one of its generating lines the line $a = 0, b = 0$, and for the tangent plane along this line the plane $a = 0$; the two quadric surfaces meet in a quartic curve passing through the two points $(a = 0, b = 0, 3ce - 2d^2 = 0)$, which are points on the torse; it is to be shown that each of these points counts three times among the intersections of the torse with the quartic curve, the number of the remaining intersections being therefore $24 - 6 = 18$; and in order thereto it is to be shown that each of the points is

[End of transcription for pages 201–250]